Examiner only

Answer **all** questions.

1. (a) (i) The box below contains the names of some common types of smart material.

photochromic pigment	**hydrogel**	**shape memory alloy**
shape memory polymer	**thermochromic pigment**	

Choose from the box the type of smart material that is used in:

I. disposable nappies, [1]

....

II. lenses for sunglasses. [1]

....

(ii) What must be done to a thermochromic pigment for it to change colour? [1]

Tick (✓) the box next to the correct answer.

heat it ☐

add water ☐

place it in sunlight ☐

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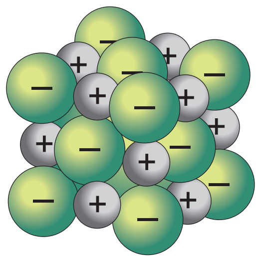
(b) (i) The following diagram shows a giant ionic structure.



Giant ionic structures have high melting points and conduct electricity when dissolved or molten.

Use the diagram and your knowledge to draw **one** line from each property to its correct explanation. [2]

Property	**Explanation**
	ions are regularly arranged
high melting point	strong bonds between ions
	ions are free to move
conducts electricity when dissolved or molten	ions cannot move
	weak bonds between ions

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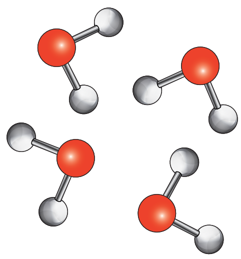
(ii) The following diagram shows a simple covalent structure.



Simple covalent structures have low melting points and do not conduct electricity.

Use the diagram and your knowledge to draw **one** line from each property to its correct explanation. [2]

Property	**Explanation**
	strong forces between molecules
low melting point	molecules cannot move
	weak forces between molecules
does not conduct electricity	molecules are not charged
	molecules are tightly packed

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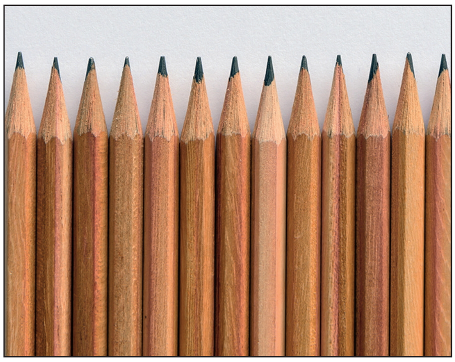
(c) Diamond and graphite have giant covalent structures. Diamond is commonly used in drill tips, whereas graphite is used in pencils.

drill tips



pencils

Use your knowledge of the properties of diamond and graphite to underline the correct word in the brackets to complete each sentence. [2]

Diamond is used in drill tips because it is (**hard / soft / malleable**) and has a high melting point.

Graphite is used in pencils because the layers of atoms are held together weakly and are able to slide (**through / into / over**) each other.

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